

Choosing a Right-Triangle Method

High School Geometry

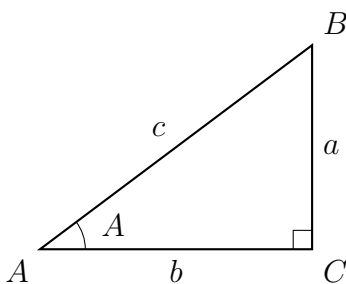
Decide first: two sides means Pythagoras, a 45 or 30 degree angle means a special triangle, anything else means a trig ratio.

Every problem below is solved with right-triangle reasoning, but they do not all want the same tool. Look at *what is given* before you write anything:

- **Two sides, find the third** — Pythagorean theorem ($a^2 + b^2 = c^2$). No angle needed and none should be used.
- **A 30°, 45°, or 60° acute angle** — special-triangle ratios give an exact answer with no calculator.
- **Any other angle with a side, or two sides and a missing angle** — a trig ratio (SOH-CAH-TOA), or an inverse trig ratio when the angle is what is missing.

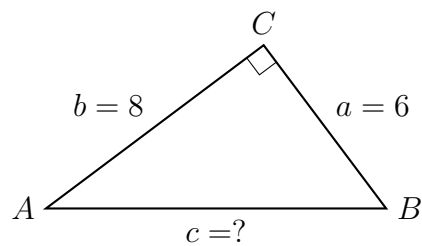
On every problem, name the method you chose before you compute, and round lengths and angle measures to two decimal places unless the answer is exact.

How every triangle here is labelled. No values are shown below — use the numbers given in each problem's own figure.



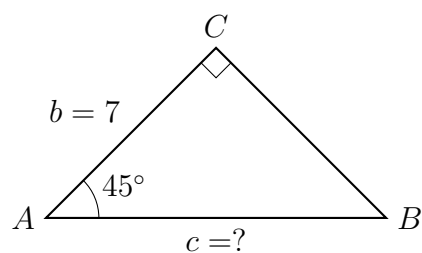
How every triangle here is labelled. No values shown: use the numbers given in each problem.

1. Find the length of side c .



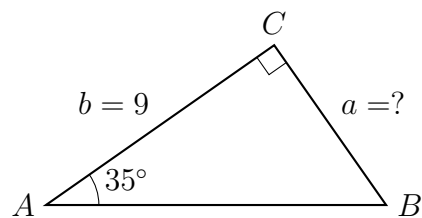
Answer: _____

2. Find the length of side c .



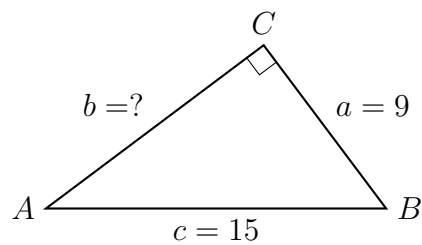
Answer: _____

3. Find the length of side a .



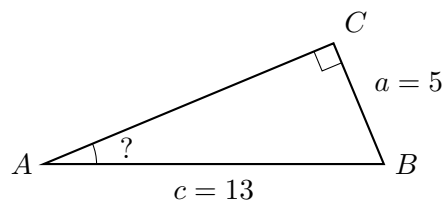
Answer: _____

4. Find the length of side b .



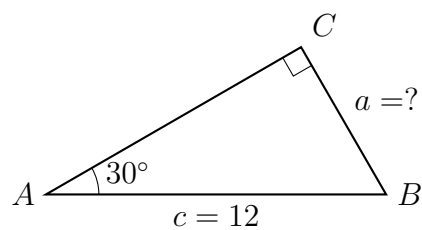
Answer: _____

5. Find the measure of angle A .



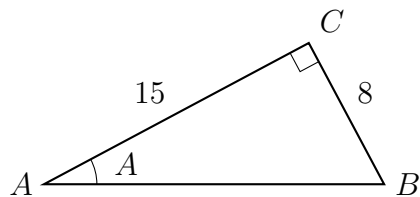
Answer: _____

6. Find the length of side a . Give an exact answer — this one needs no calculator.



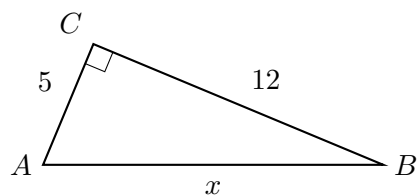
Answer: _____

7. Write $\tan A$ as a fraction in lowest terms.



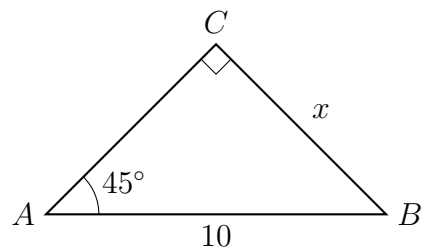
Answer: _____

8. A ladder leans against a vertical wall. The foot of the ladder is 5 ft from the wall and the top of the ladder reaches 12 ft up the wall. How long is the ladder?



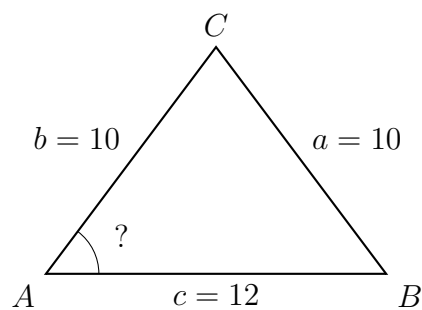
Answer: _____ ft

9. The diagonal of a square measures 10 cm. The diagonal cuts the square into two congruent right isosceles triangles, one of which is shown below. Find the side length of the square.



Answer: _____ cm

10. Challenge. The triangle below is isosceles, not right angled. Find the measure of each base angle. *Hint:* the height from C to side c splits it into two congruent right triangles — decide which method fits one of those.



Answer: _____